

G-Force Sangeet

An AI-Native Music Curriculum for Fully Autonomous Delivery on GraceandForce.com

Design mandate: Every lesson, assessment, feedback loop, and progression decision in this curriculum can be executed by the platform alone — voice AI, signal analysis, and LLM evaluation — with zero human intervention. Teachers and parents observe through the dashboard; they never grade, unlock, or intervene.

1. Design Philosophy

Music is uniquely suited to machine delivery because its core skills are **measurable signals**: pitch is frequency, rhythm is timing, dynamics is amplitude. Where debate required an AI judge of arguments, music allows something stronger — objective scoring of the student's actual sound, plus AI-rubric scoring of creative work.

The curriculum follows three spines:

1. **Dual-tradition foundation** – Indian classical (swara, tala, raga) and Western (solfège, staff notation, chords) taught in parallel, because the Indian K–12 learner deserves both vocabularies.
2. **The 4C mapping** – Creativity (composition), Critical Thinking (ear training and analysis), Communication (performance and expression), Collaboration (asynchronous ensemble challenges and cohort battles).
3. **Grammar alignment** – Music sits at the intersection of the Somatic, Mathematical, and Narrative grammars: the body produces it, mathematics structures it, story gives it meaning.

The autonomy rule: No content enters the curriculum unless one of these five machine-assessment methods can grade it:

Method	What it grades	Engine
Pitch analysis	Singing/humming accuracy against target	Pitch-tracking (autocorrelation / CREPE-class model) on mic input
Onset/timing analysis	Rhythm claps, taps, tala keeping	Onset detection vs. click track, $\pm$ ms tolerance bands

Method	What it grades	Engine
Structured response	Theory, notation reading, ear-training MCQ	Deterministic answer key
AI-rubric evaluation	Compositions, improvisations, reflections	LLM scoring against published rubric (Claude API)
Streak/behavioral telemetry	Practice consistency, listening logs	Platform analytics

2. Platform Mapping (reuse what G-Force already has)

Existing G-Force asset	Music adaptation
ElevenLabs voice engine	Guru Voice – the AI music mentor that speaks every lesson, sings reference phrases (via TTS + pre-rendered audio), and delivers feedback conversationally

Existing G-Force asset	Music adaptation
Daily 365-day guidebook structure	365-day Sangeet Path: one 12–18 minute session per day
G-Force Coins + Reward Store	Coins earned per lesson, streak bonus, battle wins; redemption for instrument-fund vouchers, masterclass recordings, digital badges
Space Force rank ladder	Raga Ranks: Shruti → Swara → Alankar → Tala Sena → Raga Yodha → Bandish Commander → Vaggeyakara (composer-poet, the summit)
Battle events (MUN/Extempore)	Monthly Swar Yudh (Voice Battle) and Tala Clash (rhythm battle) – fully auto-judged from audio submissions
Parent Dashboard (React)	Weekly auto-generated progress narrative: pitch-accuracy trend, rhythm precision %, streak, rank movement

Existing G-Force asset	Music adaptation
G1–G12 knowledge maps	G1–G12 music knowledge maps (HTML) as the reference layer

3. The Curriculum Ladder

Four stages, twelve grades, each grade = ~90 daily sessions of core path + optional side quests.

Stage 1 – LISTEN (G1–G3): The Ear Awakens

Goal: A child who can hear accurately before producing anything.

- Sound vs. silence, loud/soft, fast/slow, high/low discrimination games
- Sa–Pa recognition; Do–Sol solfège; call-and-response humming (pitch-scored)
- Body percussion: clap patterns in 4-beat and 3-beat cycles (onset-scored)
- Instrument timbre identification (tabla vs. mridangam vs. drum kit; violin vs. flute)
- Song bank: folk songs, rhymes in mother tongue + English, national songs

- **Exit assessment (auto):** match 5 pitches within ± 50 cents; clap a 4-beat pattern at ± 80 ms; identify 8 of 10 timbres

Stage 2 – SOUND (G4–G6): The Voice and the Pattern

Goal: Reliable pitch production and internalized rhythm cycles.

- Full sargam (Sa Re Ga Ma Pa Dha Ni) and major-scale solfège; alankars 1–15
- Teentaal and Adi tala keeping with voice bols; Western 4/4 and 3/4 counting
- Introduction to staff notation reading (treble clef, quarter/eighth notes) and swara-lipi
- Karaoke-mode singing with real-time pitch contour display and post-take score
- Ear training: interval recognition (Sa–Ga vs Sa–Pa; major 3rd vs perfect 5th)
- **Exit assessment (auto):** sing an alankar within ± 35 cents average deviation; read and clap 8 bars of notated rhythm; 80% on interval MCQ

Stage 3 – SPEAK (G7–G9): Repertoire and Expression

Goal: Perform complete pieces with expression; understand structure.

- Raga introduction: Bilawal/Shankarabharanam, Yaman/Kalyani, Bhairav — aroha/avaroha, pakad, one bandish each
- Western: major/minor keys, primary chords (I–IV–V), simple chord-melody relationships
- Dynamics, phrasing, and ornamentation (meend, kan-swar / slides, vibrato) — scored by pitch-contour smoothness metrics + AI-rubric on expression
- Song production basics: recording a clean take, layering a drone/backing track in the in-browser studio
- Music history capsules: Tyagaraja, Tansen, Bach, Beatles, film music of India — with AI-generated comprehension checks
- **Exit assessment (auto):** perform one full bandish/song scored on pitch + timing + AI-expression rubric (composite ≥ 70)

Stage 4 — CREATE (G10–G12): The Vaggeyakara Track

Goal: Original composition, arrangement, and a public portfolio.

- Melody writing within a raga/key; lyric setting (Malayalam/Hindi/English)
- AI-collaborative composition: student prompts, edits, and defends choices against a generated

draft – the *defense* is what the AI rubric grades, preventing pure AI outsourcing

- Arrangement in browser DAW: intro–verse–chorus / sthayi–antara structures
 - Music entrepreneurship capsule: publishing, streaming economics, the Orange Economy
 - Capstone: a 3-track EP + a spoken "composer's statement" (voice-scored for articulation via existing G-Force speech engine)
 - **Exit assessment (auto)**: capstone scored on originality-similarity check, rubric evaluation, and technical audio quality metrics
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4. The Daily Autonomous Session Loop (12–18 min)

Every day, every grade, the same machine-run loop:

1. **Guru Voice greeting** (30 s) – streak recognition, today's goal, one line of musical lore
2. **Warm-up** (2 min) – breath + sargam or rhythm tap, instantly scored, must pass a low bar to proceed
3. **Micro-lesson** (4 min) – voice + visual card teaching one concept

4. **Practice rep** (5 min) – 3–5 scored attempts with per-attempt feedback ("Your Ga is 40 cents flat – slide up slightly")
5. **Challenge** (3 min) – the day's assessed item; score → coins
6. **Reflection prompt** (1 min) – one spoken sentence from the student, transcribed and sentiment-checked; flags frustration patterns for the adaptive engine
7. **Tomorrow's teaser** (15 s)

Adaptive engine rules (deterministic, no human):

- Two failed challenges in a row → auto-insert a remedial day drawn from a pre-built remediation bank
- Score > 90 for five straight days → skip-ahead offer
- Seven-day absence → Guru Voice re-onboarding call script + parent dashboard nudge

5. Events, Battles, and Collaboration (all auto-judged)

- **Swar Yudh (monthly)**: students submit a 60-second vocal take on a set piece; leaderboard

from composite pitch/timing/expression score;
top 10 auto-featured

- **Tala Clash (monthly):** rhythm accuracy gauntlet at escalating tempos
 - **Asynchronous ensemble:** platform splits a song into parts; students record their line against a guide track; engine mixes accepted takes into a cohort "virtual choir/orchestra" video — collaboration without scheduling
 - **Season finale (termly):** cross-school ranked championship mirroring the MUN championship format
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6. Content Production Plan (build once, run forever)

Asset	Quantity	Notes
Micro-lesson scripts	~1,080 (365 × 3 difficulty variants where needed)	LLM-drafted, expert-reviewed once at authoring time
Reference audio phrases	~2,500 clips	Studio-recorded sargam/solfège/tala

Asset	Quantity	Notes
		banks; the one place human musicians are needed – <i>at build time only</i>
Backing/drone tracks	~300	Tanpura drones in 12 keys, metronome tala loops, karaoke tracks
Assessment rubrics	48 (4 per grade)	Published to students; used verbatim by the AI evaluator
Remediation bank	~200 sessions	Mapped to the most common failure signatures
G1–G12 knowledge maps	12 HTML documents	Same format as the debate maps

7. Technical Requirements Beyond Current Stack

1. **Client-side pitch detection** (Web Audio API + a lightweight model) so scoring works on low-bandwidth mobile — send scores, not raw audio, except for battle submissions
 2. **Onset-detection scoring service** for rhythm items
 3. **Audio submission pipeline** with loudness normalization before AI evaluation
 4. **Similarity checker** for the Create stage (protects against copied/AI-verbatim compositions)
 5. **Latency-calibrated tap input** — device audio latency must be measured per session with a calibration tap, or rhythm scores will be unfair
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8. What Makes This Defensible

Yoodli and AugLi judge speech; almost no one auto-judges *music performance* end-to-end for Indian K–12 with dual-tradition content. The moat is the recorded reference bank + rubric corpus + the raga-rank gamification — content and progression IP, not the AI models, which are commodities.

Suggested first milestone: ship Stage 1 (G1 only, 90 days of content) as a 100-student pilot cohort alongside the existing Cohort structure, instrument-free (voice and clap only — zero hardware barrier), and

let the telemetry from that cohort tune the tolerance
bands before scaling.